

SHUBAIL HAQUE TURZA

Department of Information & Communication Engineering · Pabna University of Science and Technology

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RESEARCH INTERESTS

Multimodal Deep Learning, Computer Vision (Medical Image Analysis, Fine-Grained Object Recognition, Hybrid CNN-Transformer Architectures), Physiological Signal Processing (EEG, Facial, Speech Analytics), Natural Language Processing.

EDUCATION

Bachelor of Science in Engineering (B.Sc. Engg.) in ICE

May 2022 – Sep 2026

Pabna University of Science and Technology (PUST)

Pabna, Bangladesh

Cumulative GPA: **3.83 / 4.00** · Department of Information and Communication Engineering

- 4-year undergraduate degree (Session: 2020–2021, Student ID: 210606) completed with honors distinction.
- Core Coursework: Deep Learning, Computer Vision, Digital Signal Processing, Algorithms, Data Structures, Database Management Systems.

Higher Secondary Certificate (H.S.C.)

2020

Saidpur Govt. Science College

Saidpur, Bangladesh

Group: Science

Secondary School Certificate (S.S.C.)

2018

Kishoreganj Multilateral High School

Nilphamari, Bangladesh

Group: Science

RESEARCH EXPERIENCE

Research Assistant

Nov 2025 – Jul 2026

Department of Information and Communication Engineering, PUST

Pabna, Bangladesh

Project: "Development of a Real-Time Multimodal Deep Learning Framework Combining EEG, Facial, and Speech Features for Detection of Stress, Depression, and Anxiety in University Students"

Supervisor: **Dr. Md. Sarwar Hosain** (Professor, Dept. of ICE, PUST)

- Engineered multimodal preprocessing pipelines synchronizing electroencephalography (EEG) signals, facial expressions, and acoustic speech data.
- Implemented cross-attention fusion networks to capture complementary physiological and behavioral markers for mental health assessment.
- Formally awarded the Certificate of Research Assistantship (Issued 06/08/2026).

PUBLICATIONS

[1] **Shubail Haque Turza**, et al., "A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images," in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, 2026.

[2] **Shubail Haque Turza**, et al., "ResNeXt50-Guided Hybrid CNN-Transformer Framework for Early Stroke Abnormality Detection: A Comparative Deep Learning Study," in *2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII)*, 2026.

[3] **Shubail Haque Turza**, et al., "Fake News Detection in Bangla: A Transformer Ensemble Framework for Robust Classification," in *2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025 (IEEE).

COMPETITIVE PROGRAMMING & HONORS

ICPC Asia Dhaka Regional Contest 2024

Dec 2024

Contestant, Team PUST_Codo_Sapiens · Hosted at Daffodil Smart City (DIU)

Dhaka, Bangladesh

- Coach: Akif Mahdi Sir | Team Members: Shubail Haque Turza, Parvez Ali, Asif Karim.
- Competed on site against premier university algorithmic programming teams across Bangladesh.

Intra-University Programming Contest (Solver Green IUPC) — 2nd Runner-Up

Jun 2023

Team codosapiens · Hosted by Solver Green in association with Edupy Academy

PUST, Bangladesh

- Secured 3rd place university-wide, solving 3 complex algorithmic problems in 149 penalty minutes.

Online Judge Profiles

Active

- CodeChef:** Rating 1476 (Division 3, 2-Star) — codechef.com/users/turzo06
- Codeforces:** Handle: *turzo06* — codeforces.com/profile/turzo06

SELECTED PROJECTS

Hybrid YOLO-EfficientNet Object Recognition Framework

2025 – 2026

PyTorch, OpenCV, YOLO, EfficientNet

- Developed a two-stage computer vision framework coupling YOLO bounding-box proposals with EfficientNet deep representations for fine-grained multi-object recognition in dense scenes.
- Published as first author in IEEE QPAIN 2026.

ResNeXt50-Transformer Hybrid for Stroke Abnormality Detection

2025 – 2026

PyTorch, Scikit-Learn, Medical Imaging

- Engineered a hybrid vision architecture fusing ResNeXt50 convolutional features with multi-head self-attention mechanisms to detect subtle acute stroke markers in clinical scans.
- Presented at PECCII 2026.

Bangla Transformer Ensemble for Misinformation Detection

2024 – 2025

HuggingFace, PyTorch, BanglaBERT

- Constructed an ensemble classifier combining fine-tuned BanglaBERT and multilingual transformers for robust classification across low-resource Bangla linguistic corpora.
- Published in 28th IEEE ICCIT 2025.

CERTIFICATIONS

SQL (Intermediate) Skill Certification — HackerRank

Oct 12, 2024

Certificate ID: 7B31E881153C · Verified: hackerrank.com/certificates/7b31e881153c

TECHNICAL SKILLS

Programming:

Python, C, C++, Java, JavaScript, PHP, SQL, MATLAB

AI / Deep Learning:

PyTorch, TensorFlow, OpenCV, Scikit-Learn, YOLO, Vision Transformers, Multimodal Fusion

Systems & Tools:

Jupyter Notebook, Kaggle, Visual Studio Code, SQL Server Management Studio (SSMS), Git, GitHub, LaTeX, Linux

Hardware & Simulation:

Proteus, mikroC, Cisco Packet Tracer

ACADEMIC REFERENCES

Dr. Md. Sarwar Hosain

Professor & Research Supervisor

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Dr. Md. Omar Faruk

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